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**(SIXTH SEMESTER DIPLOMA EXAMINATION IN ELECTRICAL AND
ELECTRONICS ENGINEERING-----MARCH 2013**

**MODEL QUESTION PAPER
INDUSTRIAL AUTOMATION**

Time: 3hours.

(Maximum marks: 100)

PART—A

I.(Answer the following questions in one or two sentences. Each question carries 2 marks.)

5x2=10

1. State home automation
2. Define open loop controller
- 3.What is a contactor in automation.
4. Name some applications of potentiometer
5. Define a PID Controller

PART-B

II (Answer any five of the following. Each question carries 6 marks)

5x6=30

1. Write the importance of automation In terms of reliability and precision
2. Explain the control system
3. Choose a suitable 3 pole contactor for supplying a 5 kW induction load
- 4.Explain the working of a pull type solenoid
5. Draw the symbols of the following
 - a) Circuit interrupter b)Foot switch c)Liquid level switch d)Pilot lamp
 - e)3 ϕ squirrel cage motor f)Annunciator
- 6 .Explain the synchros as an error detector
7. Recognize various architectures of SCADA

PART—C

(Answer one full question from each unit. Each question carries 15 marks)

4x15=60

MODULE—I

III .a) Draw the block diagram of closed loop control system .

7

b)Demonstrate how controlled variable is kept at set point in a closed loop control system 8

OR

- IV .a)Compare advantages of closed loop control system with open loop system 7
- b)Develop the mimic diagram of a temperature controller 8

MODULE—II

- V . a)Propose a suitable circuit diagram for Hoist Control 7
- b)Explain the working principle of bimetallic type temperature sensor 8

OR

- VI. a)Draw a relay (110v) control circuit for controlling the 415 V AC motor which drives a Simple conveyor system having following controls and facilities 1. ON-OFF control Separate pushbutton 2. Overload control 3. Pilot lamp indication on tripping the motor 7
- b) Discuss the working of a control valve 8

MODULE—III

- VII a) Compare the stepper motor& servomotor mechanism 7
- b) What is meant by control transformer in servomechanism 8

OR

- VIII a)Explain the working of synchros 7
- b)compare the microprocessor and micro controller 8

MODULE-IV

- IX a)Explain pneumatic PID controller 7
- b) What is RTU? Explain its functions. 8

OR

- X a)Explain DCS and its merits 7
- b) What are the applications of SCADA 8
